



I like figuring out **why** things behave the way they do.

Software Engineering Intern · voice, real-time audio and game systems · **currently building at Sing One Song**

+91-8630588622 jiya.23bcs10043@sst.scaler.com github.com/jiya-singhal linkedin.com/in/jiyasinghal jiya-persona.vercel.app leetcode.com/u/jiyasinghal_

**196**

merged PRs across 13 repos

**21,750**

benchmark cases behind one model swap

**23 / 23**

engine-port parity tests

**5.5% → 0.6%**

false alarms on 355 real recordings

**14 games**

one interruption-safe pause system

01 EXPERIENCE

Experience

Sing One Song

TypeScript · Phaser · Web Audio · Flutter · Swift · Python

Software Engineering Intern · Bangalore · Aug 2025 - Present

A singing app whose games grade your voice in real time. 244 authored PRs, 196 merged, across a Flutter app and a TypeScript game monorepo (13 repos); 60 went through peer review.

- **Real-time pitch engine, Dart → TypeScript.** Ported a voice-controlled game's grading engine from Flutter to the shared web runtime, landing a pluggable-engine seam first so the other games stayed untouched; proved correctness with a **23/23** pitch-frame replay harness against the live Dart engine rather than from-scratch unit tests, and kept the mobile cutover one URL parameter, so rollback is one line.
- **Interruption-safe pause across all 14 games.** Any call, lock screen, notification shade or app switch freezes every game exactly where it stands and opens a shared pause menu, in gameplay and in tutorials; a **+9k-line** cross-cutting change, merged, plus a deliberate do-not-merge test build with a per-game script so the team could feel-test it before merge.
- **Why Android singers were never credited.** Native pitch frames arrive in 160 ms bursts that a per-tick sampler thinned to ~6/s against a 12/s lock floor, so credit was mathematically unreachable. Fixed the collector, an honest mic-ready contract, and a stop-during-start deadlock in the Flutter bridge, each with a test that fails on the old code.
- **Audio reliability across the Flutter / WebView boundary.** Bounded mic auto-restart with backoff, iOS interruption recovery in Swift, and three distinct mic-loss mechanisms found by differential diagnosis: a JS channel-name collision, a stale AVAudioSession, and a leaked Android audio focus.
- **Real song transposition on Flutter web.** Shipped a granular-SOLA pitch-shift AudioWorklet so key changes alter the audio, not just the labels, with a coherence gate keeping scoring aligned to what the listener hears; closed the accompanying CORS gap after reproducing it with a real-SDK probe and writing the threat model down.
- **Tutorial rebuild under a byte-identical constraint.** Rebuilt a game's onboarding (**+11.5k / -2.9k** lines) on a teammate's in-flight branch: ended an infinite retry loop that trapped struggling singers, added practice / gameplay grading parity, and proved the scoring engine unchanged with MD5 gates; the suite grew 270 → 423 tests.
- **Renderer-vs-engine drift, measured.** Hop-offs got later the longer you played: the visual cursor free-ran and latched its lead over the engine. Faster cursor broadcasts plus a proportional convergence cut the gap **53 px → 16 px**, flat across rails, pinned by regression tests and a debug readout.
- **Voice-onboarding pipeline.** Built it from scratch (FastAPI, Firebase, retry-safe Cloud Tasks), then re-architected it as concurrent orchestration on Cloud Run with asyncio and Silero VAD silence-trim; measured end-to-end cold latency fell **57 s → 37 s** on the first flow every new user hits, with warm runs faster still.
- **Audio-quality gate.** A rule-based classifier on background dB, SNR, spectral flatness and temporal variance that decides whether a recording is usable, tuned for precision on warnings because a false alarm punishes a good singer: **5.5% → 0.6%** false-positive warnings across 355 production recordings.
- **Benchmark before the model swap.** Built a Playwright harness that tests pitch models in the real browser path on **21,750** synthetic cases (29 singers × 150 MUSAN noises × 5 noise-mix levels); validated swift-f0, 40× smaller than PESTO (389 KB vs 16 MB), at 98.8% vs 96.8% pass, and kept it as a standing regression gate.
- **Test, release and analytics infrastructure.** A headless screenshot runner that drives 13 voice games to live gameplay without a microphone (26/26 captures) and posts shots on every PR; a cross-repo tutorial-drift CI gate; HTTP Basic Auth across 20 live game sites; PostHog across ~20 games with a gameplay heartbeat, since voice-controlled runs produce no taps.
- **Shared systems and house rules.** A diagnostic-tip system where each game diagnoses its own run into one verdict feeding both its result screen and the app's setlist card (six games); a shared music-toggle package adopted by four games; the mic-safe sound rule: no audio in mic-live windows, because the speaker bleeds into pitch detection.
- **AI-assisted engineering, with rails.** Authored the team's first agent self-verification skill (render, capture, diff, iterate, then report) and a company playbook; run parallel agent worktree lanes, one issue → one branch → one PR: 28 PRs in 15 days, six of them merged playtest fixes across five games in a single day, every AI-written change behind diagnose-first and byte-identical gates.
- **Backend and audio ML (first months).** Voice Activity Detection task, a Resemblyzer-based gender-detection service, CREPE syllable-level pitch data, and a noise-robustness test framework across 30 singers.

Tradeindia

Python · FAISS · Streamlit

Product & Tech Intern · Noida · Jan 2025 - Apr 2025

- **Trust and fraud tooling.** Integrated third-party identity-verification APIs (Truecaller, IDfy, Shield) into the buyer–seller flow; ran the proof-of-concept evaluation of an automated voice bot through call-transcript analysis and ASR / script tuning.
- **Hybrid search over messy listings.** FAISS vector search plus fuzzy normalisation under a weighted scoring layer, so semantic and lexical signals cover each other's blind spots: relevance **89% → 96%** across thousands of unstructured listings.

02 PROJECTS

Projects

AI Persona · a grounded voice + chat agent

FastAPI · ChromaDB · Gemini · Vapi · Next.js · Cal.com

- RAG over this resume and 13 live-fetched GitHub repos (941 chunks, MMR retrieval) serving a streaming chat UI and phone calls, with real Cal.com slot proposal and booking as tool calls; voice first-token latency ~1.0 s in production. Live at [jiya-persona.vercel.app](#), and it is the portfolio you are reading this from.
- An LLM-as-judge eval harness over 20+ adversarial prompts written to break grounding: groundedness **0.92**, hallucination **9.1%**, adversarial groundedness 1.0; a nightly GitHub Action refreshes the corpus. House rule recorded in the repo: never inflate a number.

voicequal · open-source audio-quality library

Python · numpy / scipy · pytest · PyPI

- Real-time audio-quality assessment (SNR, spectral flatness, noise floor, HNR, a four-tier verdict with hysteresis for streaming), on PyPI as `pip install voicequal` and published with its full 200-clip VocalSet + MUSAN benchmark, not a favourable subset. v0.2.0's HNR-gated decision lifts exact-tier accuracy **46% → 55.5%** and off-by-one **82% → 89.5%**; 106 tests, CI on three Python versions.
- Ported the decision layer to Dart inside a production app with 26 parity tests matching the Python reference to ~9 significant figures, which surfaced three real port bugs.

Distributed KV-Cache cluster

Python · asyncio · raw TCP · Docker Compose · AWS

- Three-node sharded cache with hash-modulo key placement and transparent request forwarding, synchronous primary–replica replication for strong consistency, per-node TTL and bounded LRU eviction, over a custom Redis-style text protocol; load-tested on AWS at **11,820 req/s**, p99 18.8 ms, zero errors across 100k requests. I can argue the write-latency trade and where it breaks.

03 TECHNICAL SKILLS

Technical Skills

languages	TypeScript / JavaScript, Python, Dart, Java, SQL; Swift and Kotlin touchpoints for iOS / Android audio
audio & real-time	Web Audio, AudioWorklet, pitch detection (CREPE, PESTO, swift-f0), VAD, DSP (SOLA, SNR / spectral metrics), mic-safe sound design
games & ui	Phaser 3, Canvas2D, Flutter, React, Next.js, Tailwind
backend & infra	FastAPI, asyncio, Firebase (Functions, Hosting, Firestore), GCP Cloud Run, Node.js, Docker, GitHub Actions, Pulumi, Vercel, Render
testing & quality	Playwright, vitest, node:test, flutter test; parity harnesses, byte-identical diff gates, CI perf pins sized from measured variance
data & retrieval	FAISS, ChromaDB, Redis, SQLite, ONNX Runtime
ai engineering	RAG, LLM-as-judge evals, agent orchestration, Claude / Claude Code, Cursor, multi-agent review and audit workflows
core cs	Data structures and algorithms, OOP, DBMS, operating systems, networks

04 EDUCATION

Education

Scaler School of Technology

Undergraduate program in Computer Science · Bangalore · 2023 – Present · CGR 7.22

Birla Institute of Technology and Science (BITS)

B.Sc. (Hons.) in Computer Science · 2023 – Present · CGPA 7.89

05 ACHIEVEMENTS

Achievements & beyond code

- **LeetCode Knight**, top ~5% by contest rating.
- **Open source:** voicequal on PyPI; merged PRs to NixOS/nixos-summer (#51), OneBusAway docs (#140), lview (#8).
- **Senior Diploma in Bharatanatyam**, Prayag Sangeet Samiti, Allahabad. Years before I debugged distributed systems, I learned precision through rhythm.

numbers are nicer when they mean something. every one here traces to a PR, a commit, or a measurement. [jiya-persona.vercel.app](#)